

PRODUCTION AI ENGINEERING · T3 · EVIDENCE REAL VS SIMULATED

T3 Real vs Simulated

Which parts of the Human-in-the-Loop POC are real control logic, which stand in for an external system, what was injected on purpose, and what was not built.

T3

T3 · Production AI Engineering

3 min read · Published 2026-10-03 · Real vs simulated

Why this page exists

A claim about a control is only as strong as the part of the system that was real when it was tested. This page draws that line for every component of the run behind the Evidence Check. No simulated system is presented as a real integration.

Real code and control logic

These ran as written and are what the experiments test.

Component	Where	Exercised by
Approval artifact and its fields (no secrets)	<code>hitl/contracts.py</code> ApprovalArtifact	H9, every B/C scenario
Action digest: SHA-256 over canonical JSON of capability, target, arguments, preconditions, policy version, risk	<code>hitl/contracts.py</code> Action.digest	H1, H2
Approver authentication from the credential or a linked chat user; eligibility (role, request chain, deployment author); two-person quorum	<code>hitl/approvals.py</code>	H4, H6d
Expiry at decision time and at resume; escalation as a recorded transition	<code>hitl/approvals.py</code> , <code>hitl/gate.py</code>	H8
State machine with compare-and-set transitions; single-use consumption on entering REVALIDATING	<code>hitl/approvals.py</code>	H3, H7, CS
Revalidation on resume: agent enabled, runtime registered, delegation active, expiry, digest, policy version and requirement, resource state, approver eligibility; request binding	<code>hitl/gate.py</code>	H3, H5, H6
Idempotency key per approval, execution record, reconciliation after a crash	<code>hitl/gate.py</code>	H7c
Credential minted only after revalidation (an id, an audience, an expiry; no secret)	<code>hitl/gate.py</code>	H9
Hash-chained audit and the 16-question reconstruction	<code>hitl/base.py</code> , <code>hitl/scenarios.py</code>	H9, every scenario
The three approval protocols (A, B, C) and the policy-enforcing gateway they share	<code>hitl/arms.py</code> , <code>hitl/gate.py</code>	all
Policy decision point with decision ids, ALLOW / DENY / REQUIRE_APPROVAL, fail closed	<code>hitl/policy.py</code>	H1b, H6c, CS
HTTP API and approval inbox (stdlib) driven over real HTTP	<code>hitl/api.py</code>	API

Simulated

Faithful local substitutes. Each row says what the substitution means for the claims.

Real system	In the POC	What that means
Datadog	one fixed event (<code>dd-evt-88121</code>)	Ingest and de-duplication are exercised; alert quality is not
Kubernetes	<code>hitl/enterprise.py</code> : rolls back to the requested version whatever is running, like <code>kubectrl rollout undo --to-revision [38]</code> ; a rollback to the version already running changes nothing; writes accept an idempotency key; an effects ledger	The H5 and H6 stale rollbacks really changed state in the simulator. The idempotency key is a simulator feature real Kubernetes lacks; a production gateway must reconcile by reading the rollout instead
Slack / Teams approval channel	messages and button clicks as function calls; chat user ids linked to principals in <code>config/principals.yaml</code>	Mapping a click to a principal is real; Slack request signing and interactivity are not
Enterprise IdP and the identity context	<code>config/principals.yaml</code> : people, roles, the agent, its runtimes, the delegation and their status	Revocation and disabling are state flips the resume reads; token exchange and attestation (Agent Identity) are consumed as inputs, not re-implemented
Human waiting	the clock advances only when a scenario moves it	A 37-minute pause costs nothing; no durable workflow engine ran
Humans	scripted clicks, the same in every arm	Decision quality, fatigue and rubber-stamping are not measured
The production incident and the agent's reasoning	a deterministic evidence correlator and a fixed plan	No model is called; the claims are about the protocol, which must hold whatever a model proposes

Injected on purpose

Condition	Scenarios
A different or mutated action presented on resume	H1, H2
The same approval presented again, by the same workflow, a second instance and a new incident	H3
A guest click, a read-only engineer, the agent approving itself, the release author, one approver twice under a two-person rule	H4
A hotfix, a manual rollback, a revoked delegation, a disabled agent, a policy change, an approver losing the role, during the pause	H5, H6
A double click, a duplicate webhook, a worker crash after the write, two racing workers (deterministically interleaved, not threads)	H7
A deny, silence, a late approval, a resume after expiry, an escalation	H8
Approval service down, policy engine down, a rollout failure, a lost response	CS (conformance suite)

Not built

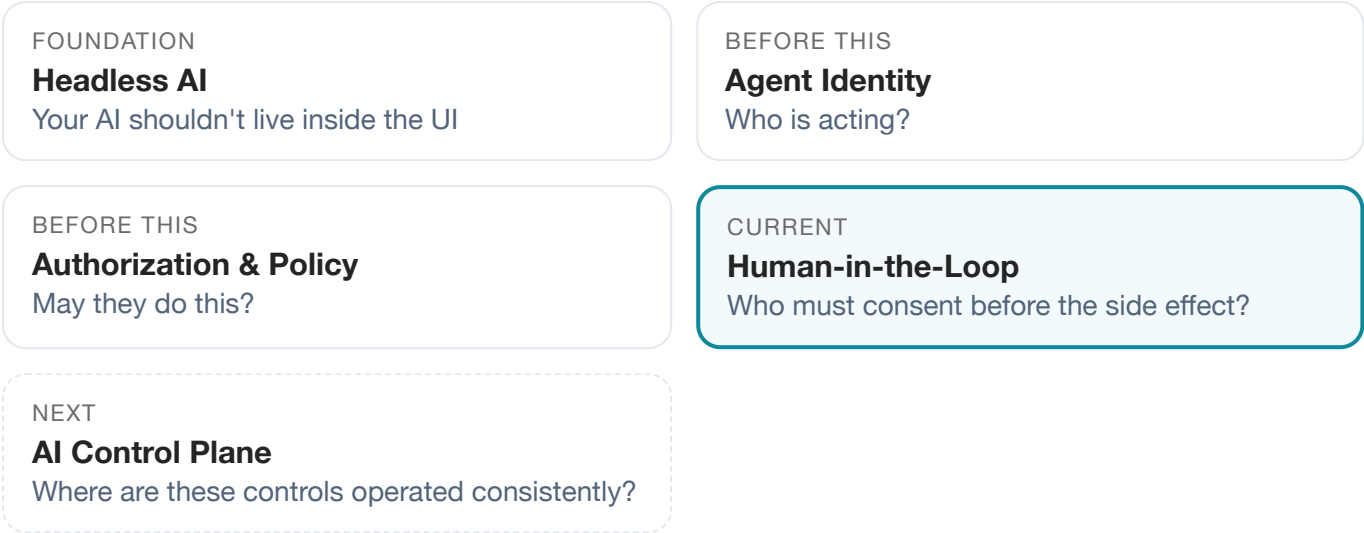
Described in the editions, not exercised: a durable workflow engine (Temporal, Step Functions), out-of-band approver authentication (OIDC CIBA ^[13]) and step-up, signed approval records, multi-agent approval chains, and the AI control plane that would run all of this for every agent.

Reproduce

```
cd hitl_poc && uv sync --group dev && uv run hitl verify 2026-10-03-protocol
```

- Evidence
- Real vs simulated
- Human-in-the-Loop

PRODUCTION AI ENGINEERING · FOUNDATION TRACK



The full learning map (Start Here)

T3 · Production AI Engineering

Production AI Engineering · T3 · Evidence. Every measured number here is substituted at build time from `hitl_poc/evidence/runs/2026-10-03-protocol/results.json` (Proof Contract v1); the enterprise systems are simulated and the run is deterministic. Figures carry their provenance.

- Medium edition Technical deep dive Evidence Check POC README Run summary
- Proof verification QA report